

What it cannot do yet

No tool solves everything, and transparency about boundaries is important. The current grammar inference engine performs best on binary protocols with fixed-length headers; text-based protocols with complex state machines (such as HTTP/2 or gRPC) show lower first-pass accuracy and may require additional training passes. Air-gapped LLM integration via Ollama is functional but slower than cloud-hosted models, and the quality of incident reports depends on the local model's capabilities. While the PQC algorithms are NIST-standardised and the platform uses *liboqs* and *oqs-rs*, the implementations have not yet undergone formal third-party cryptographic audit — contributions from the security research community on this front are especially welcome.

Why open source matters here

Security tooling that organisations cannot inspect is security tooling they cannot trust. QBITEL Bridge is released under the Apache 2.0 licence — you can read every line of the cryptography implementation, audit the ML models, and verify the claims independently. There is no open-core model and no vendor lock-in.

For India's public sector, in particular, this matters enormously. DRDO, ISRO, and defence PSUs cannot deploy cloud-dependent foreign commercial tools in classified environments. An open source, air-gappable platform built on NIST-standardised algorithms with CNSA 2.0 support provides a path to quantum readiness that proprietary tools simply cannot offer.

The community is invited to contribute across several active areas: new domain protocol modules, additional LLM provider integrations, performance benchmarks on different hardware, compliance templates for new regulatory jurisdictions, and Translation Studio language targets.

For those who would like to contribute and get involved, here are a few important details:

- Website: bridge.qbitel.com
- Repository: github.com/AXEwaves/qbitel-bridge
- Issue tracker: GitHub issues on the same repository
- Security disclosures: security@qbitel.com
- General discussion: support@qbitel.com

The quantum-safe road ahead

The convergence of quantum computing and decades of legacy infrastructure debt is not a future problem. Adversaries are already harvesting encrypted traffic. Networks are already running protocols nobody documented. The gap between what enterprise security tools protect and what actually runs on enterprise networks has never been wider.

QBITEL Bridge addresses this gap with four open, inspectable technologies working in concert: hybrid AI ensemble for protocol discovery, NIST Level 5 post-quantum cryptography for forward-secure encryption, autonomous AI agents for sub-second incident response, and Translation Studio for legacy-to-modern API conversion. The platform deploys as a transparent network layer, requires no application changes, supports nine compliance frameworks, and is fully observable through Prometheus, Grafana, and OpenTelemetry.

For any organisation running legacy mainframes, industrial control systems, or regulated workloads — this is the open source foundation for a quantum-safe future. **END** 

 By: Prabakaran Kannan

The author is VP of Technology at Innoviti Technologies, Chennai, and a research scholar in Quantum ML at NIT Puducherry. He builds open source quantum-safe security platforms.

OSFY Magazine Attractions During 2026-27

Month	Theme
March 2026	Gen AI, Deep Learning and Machine Learning
April 2026	Open Source Cloud & Infrastructure
May 2026	Blockchain and Open Source
June 2026	Modern Software Development with Open Source Programming Languages
July 2026	Observability & Performance Security & Privacy in Open Source
August 2026	All About Data Management (Database management, Optimisation, Big Data, Data analysis, etc)
September 2026	DevOps, SRE & Platform Engineering
October 2026	Open Source and Quantum Computing
November 2026	Open Source for Edge, IoT & Embedded
December 2026	Cloud Special: Everything from Management to Implementation
January 2027	Python, Data Engineering and Open Data Stacks
February 2027	Open Source Communities & Careers The Future Of Open Source